

Rotor speed at every load step

Supplement to the 5 October report, Stephen Eacuella and Taegu Kang

T. Kang's tachometer records (360 samples per second) for the eight runs were shared on 6 October. They are aligned to the rig's 1 s load steps using his generator-voltage channel alone: the two clocks differ by 0.9–1.9 s, with no measurable drift. This gives rotor speed at all 2754 load steps (43 missed pulses repaired, 12 steps excluded).

1. **The generator is the same for every rotor, at every load step.** With speed measured, $V = kn - RI + b$ gives $k = 31.3\text{--}31.5\text{ mV/rpm}$ and $R = 19.7\text{--}19.9\ \Omega$ on all eight runs. The report's inference that the gain arises in the rotor, drawn from light-load data, now holds along every load ladder. Plain's apparent source resistance in the report ($36\text{--}91\ \Omega$) is about $20\ \Omega$ of electrical resistance plus the rotor slowing under load.
2. **Power coefficient and torque** (Fig. S1). At 25.2 and 38.0 m/s, FS 0.10 and FS 0.20 reach a higher $C_{P,\text{el}}$ at a higher tip-speed ratio, and their torque-speed curves sit at higher speed. Averaged over wind speed, peak power occurs at $\lambda = 0.122$ (Plain), 0.123 (FS 0.05), 0.130 (FS 0.10) and 0.131 (FS 0.20). The generator's resistive loss barely changes the gains: +12.5% and +17.3% at the generator input, against +12.8% and +16.2% at the load.
3. **The 1 s steps are too short for full equilibrium.** Speed is still falling at the end of a step (median 0.9%; more than 2% in 15% of steps), alike for all rotors. Comparisons hold; absolute peak powers are slightly high.
4. **The generator constant fell** by 0.08 mV/rpm per hour [0.03, 0.13], 0.6% over the 2.4 h session, probably as the generator warmed. The textured rotors ran later, so this works slightly against them.

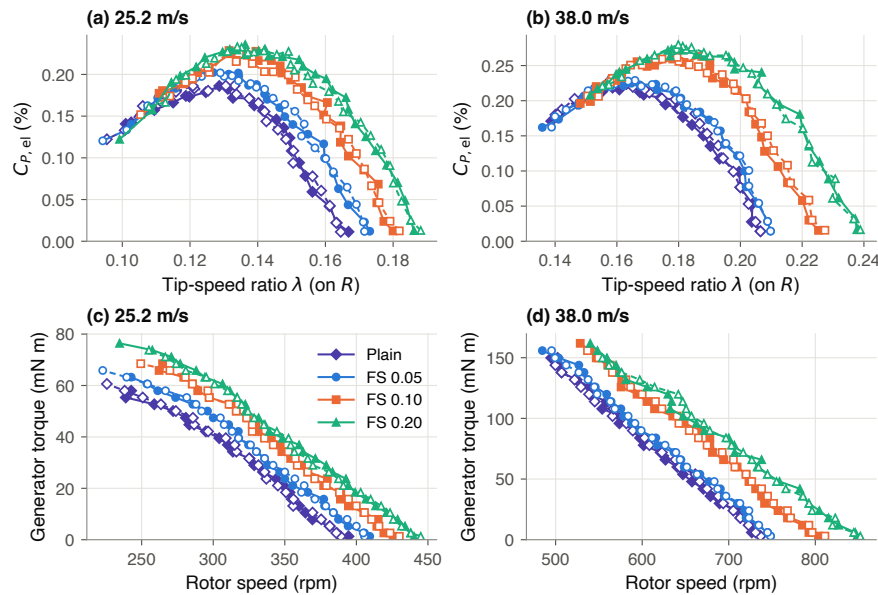


Figure S1. Electrical power coefficient against tip-speed ratio (a, b) and generator torque $k_T I$ ($k_T = 0.30\text{ N m/A}$) against rotor speed (c, d), all eight runs (solid: run 1; dashed, hollow: run 2).

Still open. The data do not yet separate more aerodynamic torque from less resisting torque: a flow effect that depends on wind speed and losses that grow with speed both fit. The torque shown is the generator's electromagnetic torque, without bearing friction or the rotor's deceleration. The tachometer magnet was re-glued for each set; only Plain and FS 0.05 lost pulses at the highest speeds, consistent with a different magnet gap. Absolute $C_{P,\text{el}}$ and λ carry the wind-calibration uncertainty. **Next round:** a redesigned base plate with a longer shaft, a balanced tachometer target, longer load steps, and an Arduino Nano 33 BLE Sense logging air conditions and vibration.